

1) Probability Table:

Label	Feature	Attribute	Frequency	Probability
yes	Shape	round	4/7	0.57
yes	Shape	square	3/7	0.43
no	Shape	round	3/8	0.375
no	Shape	square	5/8	0.625
yes	Size	large	6/7	0.86
yes	Size	small	1/7	0.14
no	Size	large	2/8	0.25
no	Size	small	6/8	0.75
yes	Color	blue	0/7	0
yes	Color	green	4/7	0.57
yes	Color	red	3/7	0.43
yes	Color	blue	3/8	0.375
yes	Color	green	2/8	0.25
yes	Color	red	3/8	0.375

2) Observation: [Square ,large, red]

Label = yes:

$$p(C) * \prod_i p(f_i|C) = \#c/\#rows * p(\text{square} | \text{yes}) * p(\text{large} | \text{yes}) * p(\text{red} | \text{yes}) = 7/15 * 0.43 * 0.86 * 0.43 = 0.074$$

Label = no:

$$p(C) * \prod_i p(f_i|C) = \#c/\#rows * p(\text{square} | \text{no}) * p(\text{large} | \text{no}) * p(\text{red} | \text{no}) = 8/15 * 0.625 * 0.25 * 0.375 = 0.031$$

$$\text{Label} = \text{yes} = 0.074 / (0.074 + 0.031) = 0.705$$

$$\text{Label} = \text{no} = 0.031 / (0.074 + 0.031) = 0.295$$